

$$1) 1. \forall x \text{ Baby}(x) \rightarrow \neg \text{Rational}(x)$$

$$2. \forall x \text{ ManageAnAlligator}(x) \rightarrow \neg \text{Loathed}(x)$$

$$3. \forall x \neg \text{Rational}(x) \rightarrow \text{Loathed}(x)$$

$$(\text{Conclusion}) \text{ Baby}(\text{Alex}) \rightarrow \neg \text{ManageAnAlligator}(\text{Alex})$$

CNF

$$1. \neg \text{Baby}(x) \vee \neg \text{Rational}(x)$$

$$2. \neg \text{ManageAnAlligator}(y) \vee \neg \text{Loathed}(y)$$

$$3. \text{Rational}(z) \vee \text{Loathed}(z)$$

$$G. \neg \text{Baby}(\text{Alex}) \vee \neg \text{ManageAnAlligator}(\text{Alex})$$

Refutation

$$\neg G. \text{Baby}(\text{Alex}) \wedge \text{ManageAnAlligator}(\text{Alex})$$

$$\neg G_1. \text{Baby}(\text{Alex}) \vee \neg \text{Loathed}(\text{Alex}) \{y/\text{Alex}\}$$

$$\neg G_2. \text{ManageAnAlligator}(\text{Alex}) \vee \neg \text{Rational}(\text{Alex}) \{x/\text{Alex}\}$$

$$(4 \text{ and } 1) \neg \text{Rational}(\text{Alex}) \{x/\text{Alex}\} / \text{Alex} \equiv \text{NIL}$$

(1 and $\neg G_1$)

$$5. \text{Loathed}(\text{Alex}) \{z/\text{Alex}\}$$

(4 and 3)

$$6. \neg \text{ManageAnAlligator}(\text{Alex}) \{y/\text{Alex}\}$$

(5 and 2)

$$7. \text{NIL}$$

$$(6 \text{ and } \neg G_2)$$